# * Assignment No:-2 *

Q.1] n = 5 (Hollow square)

(1,1) (1,2) (1,3) (1,4) (1,5)

* * * * *

(2,1)* 22 23 24 *(2,5) row → j == 1

(3,1)* 32 33 34 *(3,5) j == 5

(4,1)* 42 43 44 *(4,5) column → j == 1

(5,1)* * * * *(5,5) j == 5

(5,2) (5,3) (5,4)

```
import java.util.Scanner;
Class Demo{
        public static void main (String [] args) {
            Scanner sc = new Scanner(System.in);
            int n = sc.nextInt();

            for(int j=1; j<=n; j++){
                for (int j=1; j<=n; j++){
                    if ( j==1 || j==1 ||
                        j==5 || j==5){
                            sop ("* *");
                    }
                    else {
                            sop("  ");
                    }
                }
                sopln();
            }
        }
}
```

(int j=1 ; j<=5 ; j++) ← outer loop र तेथा

(int j=1 ; j<=5 ; j++) ← inner loop देखील 5 पर्यंत.

if (j==1 || j==5 || j==1 || j==5) ← जर j=1 किंवा 5 किंवा j=1 किंवा 5 असेल तेव्हाच * Print.

int i=1

int j=2

int j=3 ;

j=4

so all

if j=5 [j==5]

int i=1 j≤n here [j==1] j++

i≤5 cond

int j=1 j≤n j++

1≤5 cond ☑ 2

2≤5 cond ☑ 3

3≤5 cond ☑ 4

4≤5 cond ☑ 5

5≤5 cond ☑ 6

6≤5 ✗ 2

int i=2 j≤n

2≤5 int j=1 j≤n j++

1≤5 cond ☑ 2

[j==1] 2≤5 cond ☒ 3

3≤5 cond ☒ 4

4≤5 cond ☒ 5

[j==5] 5≤5 cond ☑ 6

6≤5 ✗ 3

int j=3 j≤n

3≤5 int j=1 j≤n j++

[j==1] 1≤5 ☑ 2

2≤5 ☒ 3

3≤5 ☑ 4

4≤5 ☒ 5

[j==5] 5≤5 ☑ 6

6≤5 ✗ 4

i=4 j≤n j=1 j≤n j++

4≤5 [j==1] 1≤5 ✓ 2

2≤5 ✗ 3

3≤5 ✗ 4

4≤5 ✗ 5

so all true ☑ [j==4] 5≤5 ✓ 6

6≤5 ✗ 5

i=5 [j==5] same as 2 to 4 after j=6 loop terminate.

Q.1] Diagonal square (x pattern)
If n = 5

(11) (12) (13) (14) (15)
(21) (22) (23) (24) (25)
(31) (32) (33) (34) (35)
(41) (42) (43) (44) (45)
(51) (52) (53) (54) (55)

Here i == j and
i + j = 6

i.e. $\boxed{n+1}$
n = 5 + 1 = 6

```
import java.util.Scanner;
class Demo1
        public static void main(String[] args){
            Scanner sc = new Scanner(System.in);
            int n = sc.nextInt();

            for(int i=1 ; i<=n ; i++){
                for(int j=1 ; j<=n ; j++){
                    if (i==j || (i+j)==(n+1)) {
                        sop("* _ ");
                    }
                    else {
                        sop("_ _ ");
                    }
                }
                sopln();
            }
        }
}
```

i == j i.e. 11, 22, 33, 44, 55
(i+j) == n+1 i.e. 15, 24, 33, 42, 51
6 6 6 6 6

int i=1 → j<=n
1<=5
11

j=2 j<=
2

j=3 1<=

j=4 j<=

j=5 j<=

j=6

int i=1 , i≤n

1≤5 int j=1 j≤n j++

1 1 (i==j) 1≤5 cont ☑ 2

2≤5 ☒ 3

3≤5 ☒ 4

1+5=6 4≤5 ☒ 5

(i+j==n+1) 5≤5 ☑ 6

6≤5 ✗

j++

2

j=2 i≤n

2≤5 int j=1 j≤n j++

1≤5 ☒ 2

1 2 (i==j) 2≤5 ☑ 3

3≤5 ☒ 4

2+4=6 (i+j=n+1) 4≤5 ☑ 5

5≤5 ☒ 6

6≤5 ✗

3

j=3 i≤n

3≤5 int j=1 j≤n j++

1≤5 ☒

33 (i==j) 2≤5 ☒

3+3=6 (i+j=n+1) 3≤5 ☑

4≤5 ☒

5≤5 ☒

6≤5 ✗

4

j=4 i≤4

4≤5 int j=1 j≤n j++

1≤5 ☒

4+2=6 (i+j=n+1) 2≤5 ☑

3≤5 ☒

(i==j) (4 4) 4≤5 ☑

5≤5 ☒

6≤5 ✗

5

j=5 i≤5

5≤5 int j=1 j≤n j++

1≤5 ☑

5+1=6 (i+j=n+1) 2≤5 ☒

3≤5 ☒

4≤5 ☒

(i==j)(5 5) 5≤5 ☑

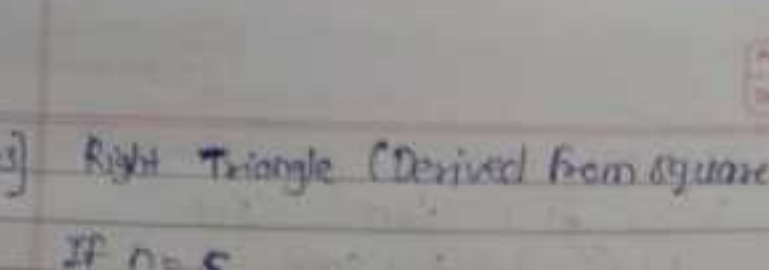
6≤5 ✗

6

j=6 i≤6 int j=1

Q.3] Right Triangle (Derived from square.)

If n = 5




if i = 1 then j = 1

```
import java.util.Scanner;
class Demo {
        public static void main (String[]args) {
            Scanner sc = new Scanner (System.in);
            int n = sc.nextInt();

            for(int i=1; i≤n; i++){
                for (int j=1 ; j≤ i; j++){
                        sop ("* ");
                }
                sopln();
            }
        }
}
```

Sol^n 1

Or

```
for(int i=1; i≤n; i++){
    for(int j=1 ; j≤n ; j++){
        if ( i ≥ j){
                sop ("* ");
        }
        else {
                sop ("  ");
        }
```

Sol^n 2

for int i=1

i=2

i=3

i=4

i=5

i=6

~~for~~ int i=1 $i \le n$

$1 \le 5$ int j=1 $j \le i$ j++

$1 \le 1$ * 2

$2 \le 1$ X

j++ 2

$i=2$ $i \le n$

$2 \le 5$ int j=1 $j \le i$ j++

$1 \le 2$ * 2

$2 \le 2$ * 3

$3 \le 2$ X

3

$i=3$ $i \le n$

$3 \le 5$ int j=1 $j \le i$ j++

$1 \le 3$ * 2

$2 \le 3$ * 3

$3 \le 3$ * 4

$4 \le 3$ X

4

$i=4$ $i \le n$

$4 \le 5$ int j=1 $j \le i$ j++

$1 \le 4$ * 2

$2 \le 4$ * 3

$3 \le 4$ * 4

$4 \le 4$ * 5

$5 \le 4$ X

5

$i=5$ $i \le n$

$5 \le 5$ int j=1 $j \le i$ j++

$1 \le 5$ * 2

$2 \le 5$ * 3

$3 \le 5$ * 4

$4 \le 5$ * 5

$5 \le 5$ * 6

$6 \le 5$ X

6

$i=6$ $6 \le n$

$6 \le 5$ X terminate the loop

Q.4] Inverted Right Triangle

If n = 5

| | | | | |
|---|---|---|---|---|
| * | * | * | * | * |
| * | * | * | * | * |
| * | * | * | * | * |
| * | * | * | * | * |
| * | * | * | * | * |

→ if i+j ≤ n+1

or

j = 5 j = 5 and j--

i.e j ≤ i

```
import java.util.Scanner;
class Demo{
        public static void main (String[]args){
            Scanner sc = new Scanner(System.in);
            int n = sc.nextInt();

            for (int i = n; i ≥ 1; i--){
                for (int j = 1; j ≤ i; j++){
                    sop("* ");
                }
                sopln();
            }
        }
}
```

or

```
for (int i = 1; i ≤ n; i++){
    for (int j = 1; j ≤ n; j++){
        if (i+j ≤ n+1){
            sop("* ");
        }
        else {
            sop("  ");
        }
    }
    sopln();
}
```

int i = 5

int i = 4

int i = 3

int i = 2

int i = 1

int i = 0

int i=5 i≥n

i≥1 int j=1 ; j≤i ; j++ j=-

1≤5 ✓ 2

2≤5 ✓ 3

3≤5 ✓ 4

4≤5 ✓ 5

5≤5 ✓ 6

6≤5 ✗ 4

int i=4 i≥n

4≥1 int j=1 ; j≤i ; j++

1≤4 ✓ 2

2≤4 ✓ 3

3≤4 ✓ 4

4≤4 ✓ 5

5≤4 ✗ 3

int i=3 i≥n

3≥1 int j=1 ; j≤i ; j++

1≤3 ✓ 2

2≤3 ✓ 3

3≤3 ✓ 4 2

4≤3 ✗

int i=2 i≥n

2≥1 int j=1 ; j≤i ; j++

1≤2 ✓ 2

2≤2 ✓ 3

3≤2 ✗ 1

int i=1 i≥n

1≥1 int j=1 ; j≤i ; j++

1≤1 ✓ 2

2≤1 ✗ 0

int i=0 i≥n

0≥1 ✗ — loop terminate.

Q.5] Chessboard Pattern

If n=4

```
*  #  *  #
#  *  #  *
*  #  *  #
#  *  #  *
```

```
if(i+j % 2 == 0){
    *
else
    #
```

```
import java.util.Scanner;
class Demo{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        for (int i=1 ; i≤n ; i++){
            for (int j=1 ; j≤n ; j++){
                if (i+j % 2 == 0){
                    sop("*_");
                }
                else{
                    sop("#_");
                }
            }
            sopln();
        }
    }
}
```

int i=1 j≤
1<
3
1
1
1
1+

int j=2 0<
2<
2+
2+2
2+3
2+4

int j=3 j
3
3
3
3
3

int j=4

int i=5

int i=1 i≤n

1≤4 — int j=1 j≤n j++ j++

1+1=2 1≤4 # 2

1+2=3 2≤4 # 3

1+3=4 3≤4 # 4

1+4=5 4≤4 # 5

5≤4 ✗

2

int i=2 i≤n

2≤4 — int j=1 j≤n j++

2+1=3 1≤4 # 2

2+2=4 2≤4 # 3

2+3=5 3≤4 # 4

2+4=6 4≤4 # 5

5≤4 ✗

3

int i=3 i≤n

3≤4 — int j=1 j≤n j++

3+1=4 1≤4 # 2

3+2=5 2≤4 # 3

3+3=6 3≤4 # 4

3+4=7 4≤4 # 5

5≤4 ✗

4

int i=4 i≤n

4≤4 — int j=1 j≤n j++

4+1=5 1≤4 # 2

4+2=6 2≤4 # 3

4+3=7 3≤4 # 4

4+4=8 4≤4 # 5

5≤4 ✗

5

int i=5 i≤n

5≤4 — loop terminated